

User Manual

Project Overview

This project aims to classify images by ethnicity using a deep learning model trained on a labeled dataset. You can either use preprocessed image data and pretrained weights or start from scratch by processing the raw dataset and training the model yourself.

Step 1: Install Necessary Requirements

Option A: Local Machine Setup

1. Create a virtual environment (recommended):

```
python -m venv venv
source venv/bin/activate
```

2. Install dependencies from requirements.txt:

```
pip install -r requirements.txt
```

Option B: Google Colab User

- If you're using Google Colab, the environment is already pre-configured with most libraries.
- You can skip installing from requirements.txt.
- Just ensure you upload necessary files and adjust paths accordingly.

Step 2: Download Required Artifacts

Download from Google Drive link below:

https://drive.google.com/drive/folders/1E2L21N_4hzEyoUNETxCZc_LI2b8OkSsB?usp=drive_link

Contents	Description
trained_model folder	Fully trained ethnicity classification model. It can be directly loaded for evaluation or inference.
test_df_preprocessed.pkl	Preprocessed image features and labels for testing.
train_df_preprocessed.pkl	Preprocessed image features and labels for training.
vgg_face_weights.h5	Pretrained VGG-Face weights used for transfer learning in the model architecture.

Step 2: (Optional) Start with Initial Image Processing

If you want to train the model from scratch and process the raw dataset manually, follow these steps:

1. Prepare the Dataset:
 - Download and unzip the raw dataset.
2. Run Preprocessing Notebook:
`Initial_Image_Processing_(Skip_If_Already_Saved_as_pkl).ipynb`

This will:

- Resize, normalize, and prepare images
- Generate .pkl files and save them inside the artifacts/ folder

Make sure you update the notebook paths based on your local folder structure.

Step 3: Train the Model

Once the .pkl files are ready (either by download or preprocessing), you can train the model.

Open and run the notebook: `Model_Training_Source_Code.ipynb`

This notebook will:

- Load the training and validation .pkl files
- Initialize the model architecture
- Train the model and save weights